BASE PAPER OVERVIEW

PixArt-α: Fast Training of Diffusion Transformer

Core Problem

Training high-quality text-to-image diffusion models is extremely slow and compute-intensive, requiring millions of GPU-hours and large datasets.

Proposed Solution

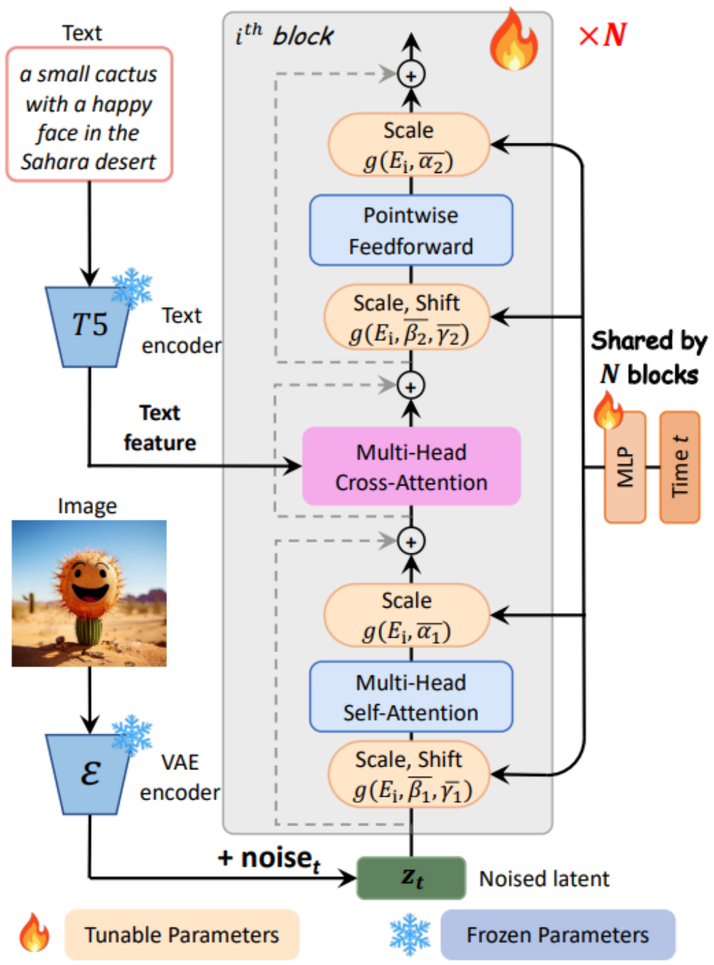
PixArt-alpha decomposes training into 3 stages: class-conditioned pretraining, concept-alignment, and high-resolution refinement via a DiT backbone.

Key Results

Achieves state-of-the-art FID 7.32 on COCO dataset at 1024x1024 resolution using only ~10% of the compute needed by comparable models like Stable Diffusion XL.

Limitations

Weak control over object count and fine details (notably hands).
It also has limited text generation capability due to lack of diverse font/character data.



Architecture: DiT (Diffusion Transformer) backbone · T5 text encoder · Multi-scale VAE · Cross-attention conditioning

OUR APPROACH & CONTRIBUTIONS

Targeted LoRA Fine-Tuning for Anatomically Correct Hands

Problem: Diffusion models consistently fail at generating anatomically correct hands - fused fingers, wrong counts, distorted geometry.

01

Custom Hands Dataset

Curated high-quality hand images with LLaVA-generated anatomical captions emphasising finger count, separation, and geometry.

02

LoRA Fine-Tuning

Applied LoRA (r=16, alpha=32) to DiT attention layers (to_q, to_k, to_v, to_out). Froze VAE & T5 encoder, only adapter weights trained.

03

Optuna HP Tuning

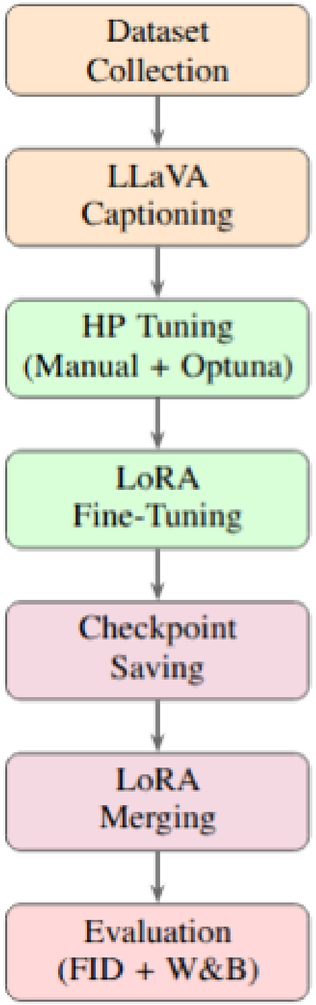
Automated search over LR, rank, alpha, dropout, scheduler.

04

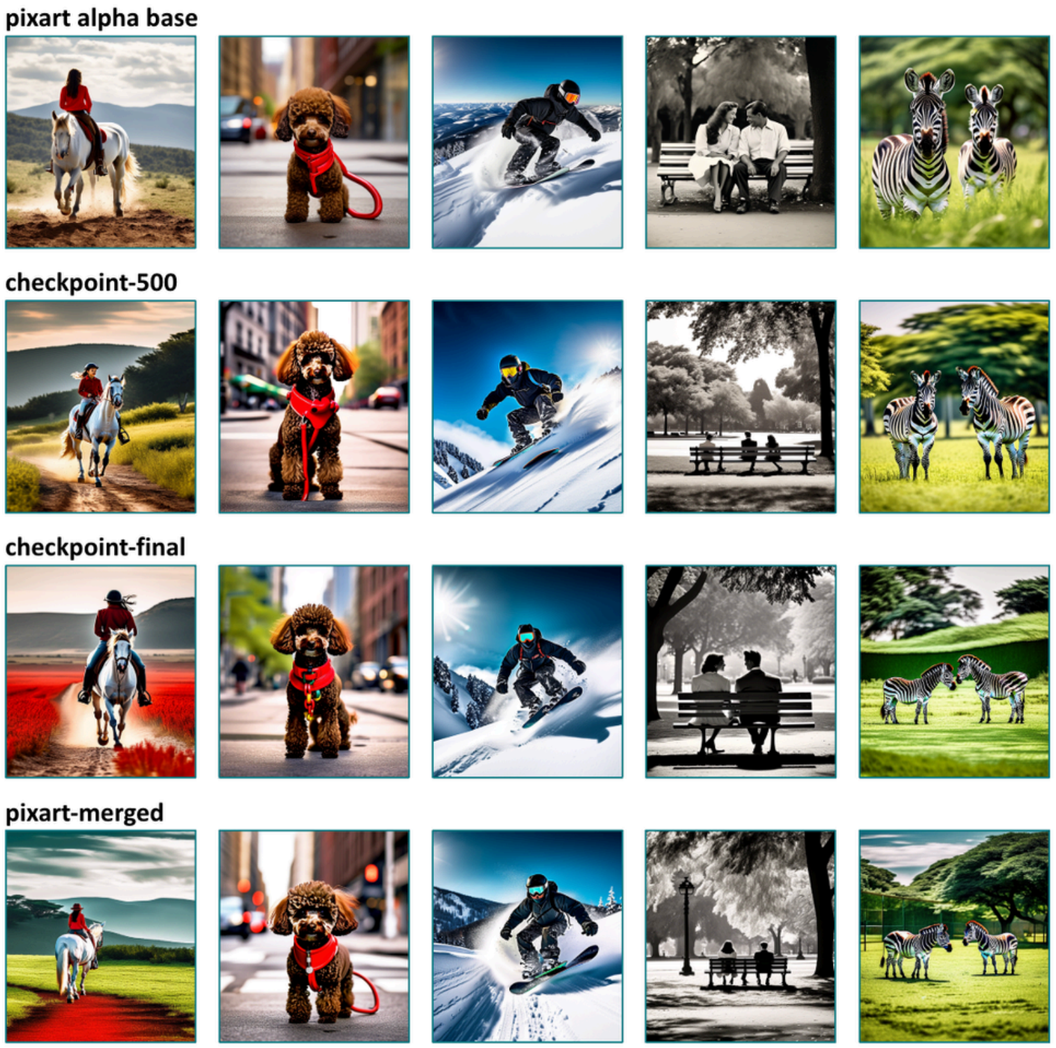
Evaluation Pipeline

W&B logging of loss curves, grad norms, LoRA weight norms, and visual samples every 100 steps. Base vs checkpoints vs merged model.

Ran multiple experiments with varied datasets (easy / normal / hard hand images) and HP configurations to find optimal settings



Quantitative Metrics + Visual Evaluation



Quantitative Results

Model	FID ↓
PixArt-alpha Base	126.73
LoRA ckpt-500	116.41
Merged Model	109.61

Fused / extra fingers

Improved separation

Best anatomy

Model	FID ↓
Learning Rate	5 x 10-6
LoRA Rank (r)	16
LoRA Alpha (α)	32
Batch Size	2
Epochs	15
Dropout	0.1
LR Scheduler	Cosine
Warmup Steps	200
Gradient Accumulation	7.5
Resolution	512

Individual Contribution

Implemented LoRA training script
Integrated Optuna HP finetuning
Integration of WandB logging
Report & Presentation
- Arpita

Dataset ratio configuration
Dataset LLaVa captioning pipeline
Implemented LoRA training script
Gradio demo & Presentation
- Prajna

Dataset LLaVa captioning pipeline
Inference scripts for FID
T2I-Compbench
Video
Report & Presentation
- Sreenitya

GitHub: PixArt-Alpha-LoRA · YouTube: youtu.be/Aoez03xao6E · HuggingFace: arpita2desh/pixart-hands-lora